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7(b)	

<table>
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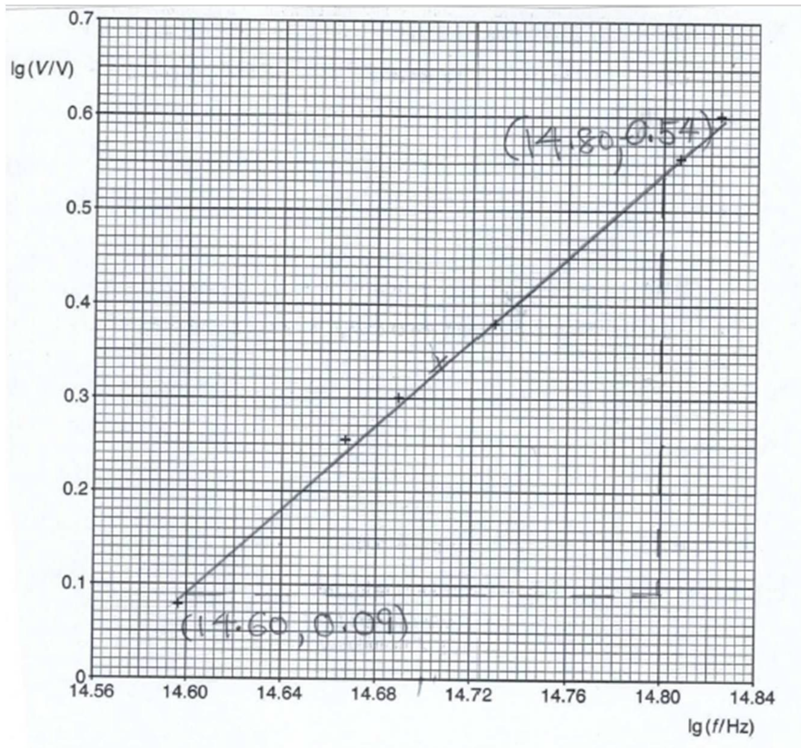
<tr><td>7(b)</td><td>


Fig. 7.1

$$n = \frac{0.54 - 0.09}{14.80 - 14.60} = 2.25$$
</td></tr>
<tr><td>7(c)(i)</td><td>
$$f = \frac{3.0 \times 10^{8}}{520 \times 10^{-9}} = 5.77 \times 10^{14} \text{ Hz}$$
lgf = 14.761

From graph, lgV=0.45 => V = 2.82 V
</td></tr>
<tr><td>7(c)(ii)</td><td>
P = nE, where P is the power, E – photon energy, n=no of photons per unit time.
$$n = \frac{10}{3.8 \times 10^{-19}} \text{ s}^{-1}$$
No of photons released in 1.0 min = $60 \times \dfrac{10}{3.8 \times 10^{-19}} = 1.58 \times 10^{21}$
</td></tr>
<tr><td>7(c)(iii)</td><td>
$V_R + V_{LED} = 4.5$
$V_R + 2.85 = 4.5$
$V_R = 1.65$ V
From passage, normal operating current through the LED is 20 mA, 0.020 A.
Hence the current passing through the resistor in series is 0.020 A.
</td></tr>
</table>

	$V_R = IR$ $R = \dfrac{1.65}{0.020} = 82.5\ \Omega$
7(d)	$Q = It$ from the concept that current is the rate of flow of charge 730 mAh means that the charge flowing for a current of 730 mA for 1 hour.
7(e)	For LED, $e_{LED} = \dfrac{800\ \text{lumen}}{10\ \text{W}}$ For incandescent lamp, $e_{lamp} == \dfrac{840\ \text{lumen}}{60\ \text{W}}$ $\dfrac{e_{LED}}{e_{lamp}} = \dfrac{\frac{800}{10}}{\frac{840}{60}} = 5.71$
7(f)(i)	To get similar/same brightness for 50 000 hours of use, LED lamp costs \$35.95 CFL lamps cost 5 × \$3.95 = \$19.75 Incandescent lamps cost 42 × \$1.25 =\$52.50 Hence from the cost and frequency of replacement, LED and CFL , people would prefer LED and CFL.
7(f)(ii)	Total cost of using one LED traffic light for 50,000 h $= 35.95 + (0.22 \times \dfrac{50000 \times 10}{1000})$ $= \$145.95$ Total cost of using one CFL traffic light for 50,000 h $= (3.95 \times 5) + (0.22 \times \dfrac{50000 \times 14}{1000})$ $= \$173.75$ The total cost of using LED traffic light is less than that of CFL traffic light. Furthermore, CLF needs 5 replacements to achieve 50,000 operating condition, in which the labour cost will further add on to the total cost. Hence LTA's decision to replace all the CFLs used in traffic lights with LEDs is justifiable.
7(f)(iii)	Incandescents offer reliable, even lighting without flickering. Unlike some types of lamp such as CFLs, incandescents do not have a warm up period as you flick the switch. This ensure quick response in traffic lights.